Executive Summary - Stadtbibliothek Data Analysis

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Introduction: This project investigates transaction data from the library management system of the Munich Public Library (Stadtbibliothek München) for the 4th quarter of 2025. The dataset contains information on relevant library branches, time, genre and user characteristics, such as age and gender.

Research Objectives: The main goal is to explore how library services are utilized across different branches, time periods, and user characteristics. The analysis consists of three main parts. First, the spatial and structural perspective examines how transactions are distributed across the branches and identifies the relation between them. Second, the temporal part explores how usage patterns vary monthly, weekly, and daily. Third, the user perspective aims to answer how the transactions are distributed across the genres and how these preferences are associated with age and gender.

Challenge and Methodology: The data preprocessing was substantial due to a high percentage of invalid lending library identifiers (~83%), resulting in a large reduction affecting the spatial analysis. To address this, entries with invalid lending library ID were removed for the spatial part and the data was structured depending on the specific research questions, with different subsets used for spatial, temporal, and user-focused analyses. The project applies descriptive and comparative methods, including aggregation by library, time, and user characteristics. Spatial patterns were examined through the comparison of own-branch and inter-branch transactions, while temporal trends were analyzed across months, weeks, weekdays, and hours. User behavior was analyzed by comparing genre preferences and examining how they differ across age and gender.

Key Findings and Conclusion: The analysis reveals clear spatial, temporal, and demographic patterns in library usage. Libraries located in central areas show a larger number of transactions, with 3 branches (Motorama, HP8, and the Central Magazin) acting as key hubs for the inter-branch transactions (~ 32%), while other hubs are more broadly distributed. Usage patterns exhibit certain temporal regularities, with peak activity observed on Tuesdays, particularly in the afternoon (3-5pm), and a noticeable decline during the Christmas period. In terms of user behavior, children aged 7 to 12 represent the most active group, highlighting the importance of youth engagement within the library system. Furthermore, female users account for a higher number of transactions across almost all genres (except for comics), and from around age 20 onwards, women consistently use the library more than men. Additionally, clear age-specific preferences are observed, with children predominantly borrowing comics and youth literature, while literature and fiction become increasingly popular among older users. These findings indicate that spatial, temporal, and demographic factors jointly shape library usage patterns.

Outlook: The analysis provides valuable insights into library usage patterns and user behavior within the Munich Public Library system. Future analyses could benefit from more complete and consistent data, particularly regarding lending library ID, as well as an extended period of observation beyond a single quarter to capture long-term trends. The identified spatial, temporal, and demographic patterns may offer a foundation for optimizing resource allocation, staffing, and collection management across library branches. Besides, the results can support data-driven decision-making for the management of the Munich Public Library system.